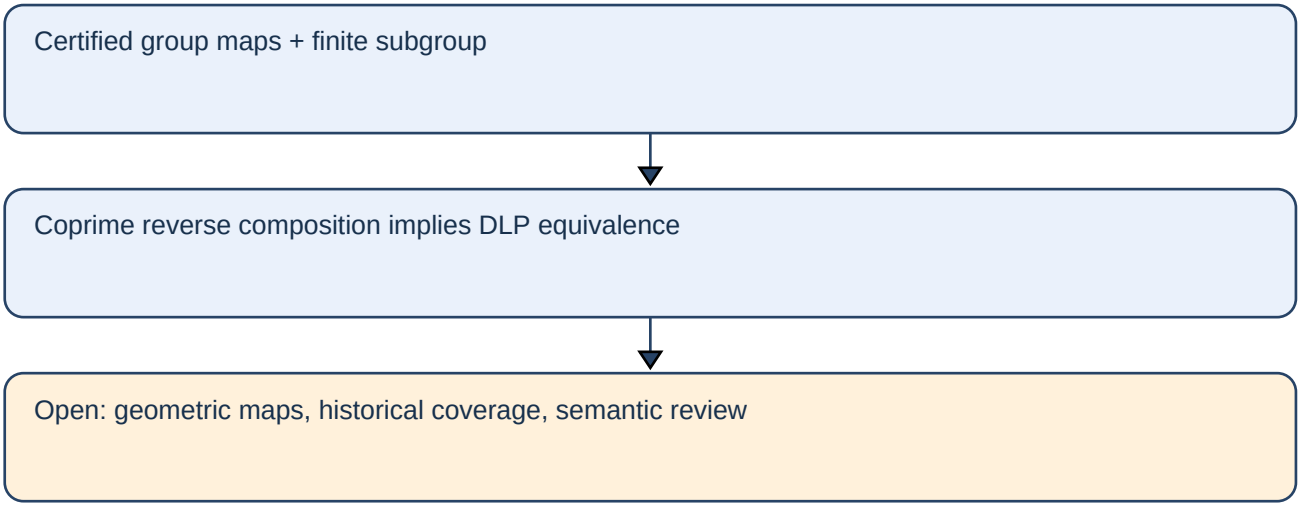
crypto: checked statements and open bridges



Logical obligations only. No curve route or measured performance is represented.

Lean retrospective, 2026-10-18

Status: **partial retrospective; five new theorems machine checked**. Independent semantic review is pending. This does not certify every historical claim in crypto, cryptanalysis, or crypto-autoresearcher.

Request and coverage

The requested outcome is a retroactive review and faithful Lean proofs of the past mathematical claims across all three repositories. This addition starts that review without changing frozen reports, measurements, or scientific state. The baseline is crypto commit `abee07ee7337d04a7e734333669ca1bacd6c763a`.

Requirement	Status	Evidence or remaining work
Inventory existing formal source and projects	Complete for this Git tree	inventory.json: 98 Lean source paths excluding self-test fixtures, 19 Lake projects; these are path counts, not verified-claim counts
Re-prove transfer using actual group laws	Machine checked, conditional	Transport.lean, proof map
State transfer on elliptic-curve points	Machine checked, conditional	`curve_dlog_transport_iff`, using Mathlib `WeierstrassCurve.Affine.Point`
Re-prove factor-base range containment	Machine checked	`closure_le_range`, `outside_range_not_in_closure`
Recheck all historical crypto proofs and prose claims	Open	The inventory is not a build or a complete extraction of prose claims
Independent semantic-fidelity review	Pending	Author correspondence notes below are not an independent review

Statements, proofs, and correspondence

The historical sources are TransferBound.lean and SubfieldCollapse.lean.

For a finite subgroup S, additive homomorphisms f and back, and degree d, assume `back(f(P)) = d • P` for every P in S and `gcd(|S|,d)=1`. If f(P)=f(Q), applying back gives dP=dQ. Multiplication by d is bijective on S, so P=Q. Consequently `k • f(G)=f(Q)` iff `k • G=Q`, for every integer k. The curve theorem specializes these exact hypotheses to actual elliptic-curve point groups. It does not construct an isogeny, its reverse map, or the duality identity, and does not establish a measured cost or a conductor-dependent gain. Those bridges remain necessary for any concrete curve claim.

For an additive homomorphism f, its range is an additive subgroup. If every factor-base element lies in that range, the universal property of subgroup closure puts the entire generated subgroup in the range. A target outside the range

therefore cannot be generated by that base. This version uses Mathlib groups and closure. Construction of a Jacobian base-change map and a Weil cardinality bound remain outside this theorem.

Reproduction and evidence

Lean `v4.35.0`; CryptoLean commit `68117089795313bc2ffe3a3b5f72052d5fa7cb91`; transitive dependencies are pinned in lake-manifest.json. Local builds reused checkouts of those exact revisions through ignored `.lake/packages` symlinks.

```
bash research/lean_retrospective_20261018/check.sh
```

verification.log retains the build, every `#print axioms`, and the fail-closed transitive axiom audit. All five theorems use only `propext`, `Classical.choice`, and `Quot.sound`; zero compiler-trusted proofs. The existing repository-wide Lean workflow discovers this Lake project. `lean4checker` is not installed, so fresh external rechecking was not performed. The requested Rust check could not launch because `cargo` is not installed; no Rust source is changed. `conductor` is also unavailable; the implementation is confined to this new study and the problem-index pointer.

Reproduce the baseline path inventory, using a fresh output filename:

```
git ls-tree -r abee07ee7337d04a7e734333669ca1bacd6c763a | node
research/lean_retrospective_20261018/inventory.mjs abee07ee7337d04a7e734333669ca1bacd6c763a /tmp/crypto-
inventory.json
```

Visual record and open work

Diagram, editable diagram data, and report PDF summarize the proof chain. No new curve, subgroup instance, isogeny edge, benchmark value, or speedup is asserted. The source graphs `docs/ic/boundary_targets.json`, `docs/ic/progress-timeline.json`, `docs/index-calculus-scoreboard.html`, and `docs/ic-leaderboard.html` were checked for these source-study references; no result row needs changing. The diagram describes logical obligations, not a measured curve route.

The cross-repository coverage gap is tracked in the canonical problem record. Next obligations include the concrete geometric maps, a claim-by-claim prose inventory, independent semantic review, and builds of the remaining historical crypto projects. No timeout or missing proof establishes mathematical falsity.